Redox

Documentation Infrastructure Audit Report

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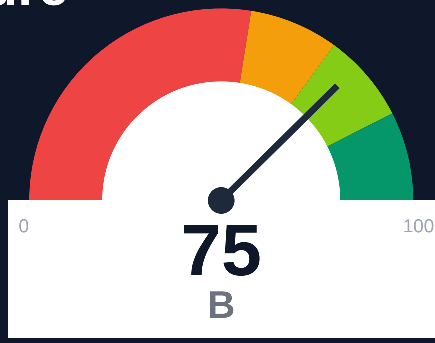


Risk: Moderate

Key Metrics

Pages scanned: 1315

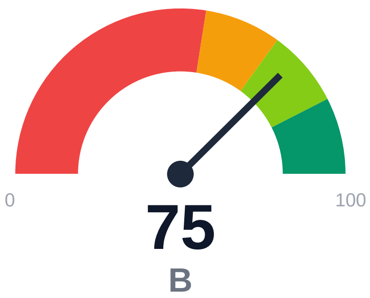
Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 679
- SEO/GEO issue rate is high: 29.11%
- Example reliability estimate is low: 55.51%

Executive Summary



Documentation audit of docs.redoxengine.com crawled 1,315 pages from 1,994 discovered URLs with 100% crawl coverage. Critical issue: 679 confirmed broken internal documentation links affecting user navigation. Example code reliability estimated at only 55.51%, indicating potential execution or maintenance issues. SEO/geo metadata issues affect 29.11% of pages, and last-updated timestamps are missing on 40% of content.

External posture: SEO/GEO issue rate **29.1%**, public API coverage **100.0%**, broken links **679**.

75 Audit Score	**B** Grade	**Moderate** Risk Band
1315 Pages Scanned	**679** Broken Links	**29.1%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://docs.redoxengine.com/	1315	679	100.0	55.5	29.1

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-17**
Industry average (developer docs)	**85**	**-10**
Minimum acceptable	**70**	**+5**
Your score	**75**	**-**

Gap to industry average: **10 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public)
100.0%

SEO/GEO Issue Rate (lower is better)
29.1%

Example Reliability (estimate)
55.5%

Freshness Visibility (last-updated metadata)
60.4%

Internal Metrics (from scorecard)

API Coverage (Internal)
0.0%

Example Reliability (Internal)
100.0%

Docs-Contract Drift
0.0%

Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
$140,822 annual exposure	**$249,513** annual exposure	**$431,728** annual exposure
Remediation: **$2,090** Monthly loss: **$4,125** Opportunity: **$7,436**/mo	Remediation: **$5,445** Monthly loss: **$12,903** Opportunity: **$7,436**/mo	Remediation: **$8,800** Monthly loss: **$27,808** Opportunity: **$7,436**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	$1,782	$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	$1,122	$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	$2,257	$990
F-API-COVERAGE	Undocumented API operations	LOW	$2,904	$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	$2,244	$935
F-DRIFT	Code/docs contract drift	LOW	$1,901	$660
F-TERMINOLOGY	Terminology inconsistency	LOW	$693	$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Audit and repair all 679 broken internal documentation links to restore navigation integrity [impact: critical, effort: high]

2. Review and update code examples to improve 55.51% reliability score; implement automated testing for examples [impact: critical, effort: high]

3. Add or update last-updated metadata on 40% of pages missing freshness indicators [impact: high, effort: medium]

4. Resolve SEO/geo metadata issues affecting 29.11% of pages to improve search discoverability [impact: high, effort: medium]

5. Investigate 107 trap URLs and 687 uncrawled pages to identify structural or access issues [impact: medium, effort: low]

Expert Analysis: Strengths and Risks

Strengths

+ Complete crawl coverage at 100% with 1,315 pages successfully indexed
+ API reference documentation detected on 848 pages with 100% reference coverage
+ Overall audit confidence level of 91.5% indicates reliable findings
+ Single-product topology provides clear documentation structure
+ No repository navigation link issues (0 repo broken links)

Risks

- 679 confirmed broken internal documentation links create severe navigation barriers for users
- Example code reliability at 55.51% suggests nearly half of code samples may be outdated or non-functional
- 29.11% SEO/geo issue rate impacts discoverability and search engine ranking
- 39.62% of pages lack last-updated timestamps, obscuring content freshness for users
- 107 trap URLs were skipped, indicating potential crawl obstacles or infinite loop patterns
- API confidence at 85% (lowest subscore) suggests potential gaps in API documentation detection
- 687 URLs requested but not crawled may indicate access issues or redirect problems

Limitations

* External audit cannot verify actual functionality of API endpoints or validate that code examples execute successfully in live environments
* Broken link detection reflects point-in-time status; links may break or be fixed after audit completion
* Example reliability is estimated through heuristics; actual code correctness requires manual review and testing
* Audit cannot assess content accuracy, completeness of technical explanations, or alignment with current product features
* SEO/geo scoring reflects metadata presence and structure but cannot predict actual search engine ranking performance

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.